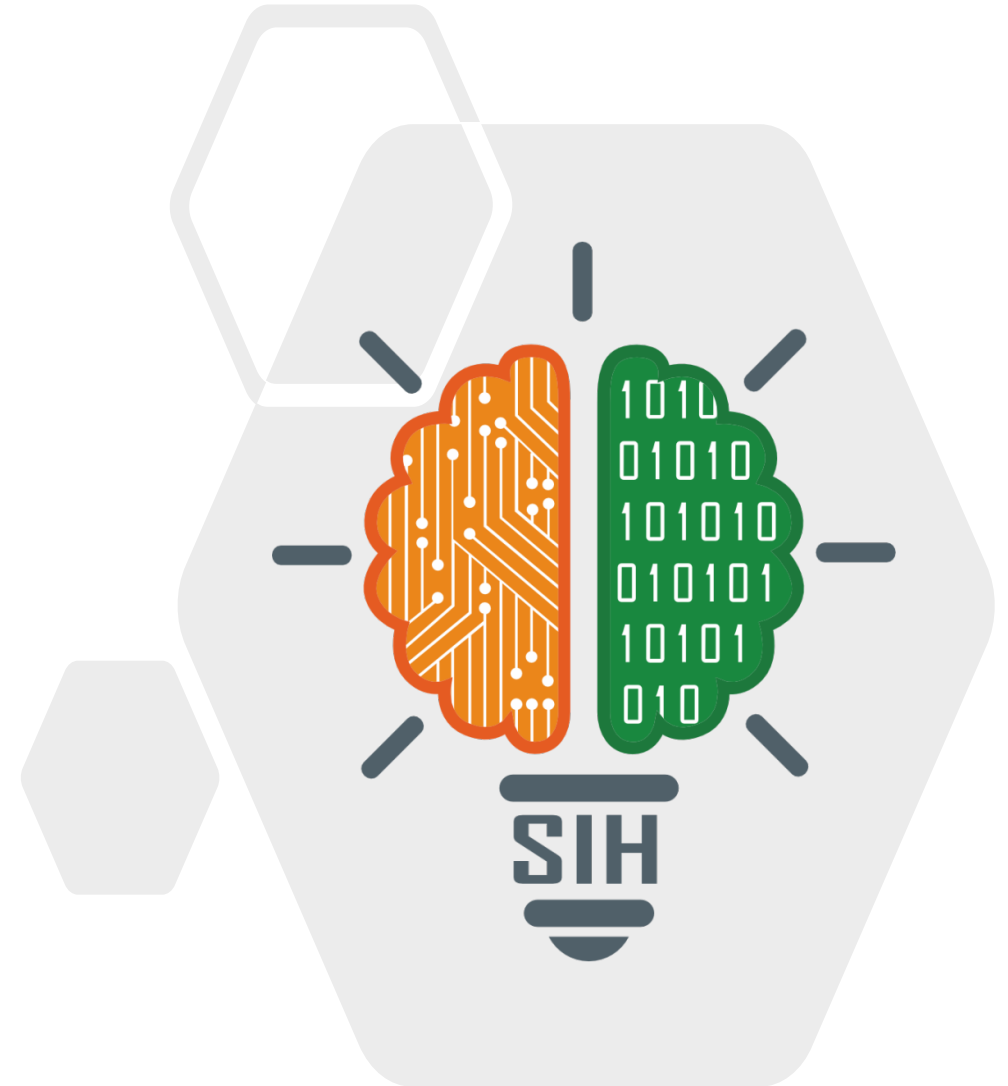


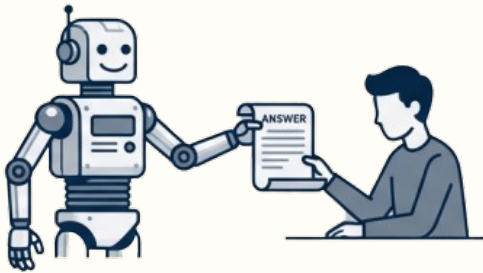
TITLE PAGE

- PROBLEM STATEMENT ID: **N/A**
- PROBLEM STATEMENT TITLE:
**POST-CLASS LEARNING SUPPORT
PLATFORM**
- THEME: **EDUCATION**
- PS CATEGORY: **SOFTWARE**
- TEAM ID: **N/A**
- TEAM NAME: **NOHARA FAMILY**



SHIKSHA SAATHI

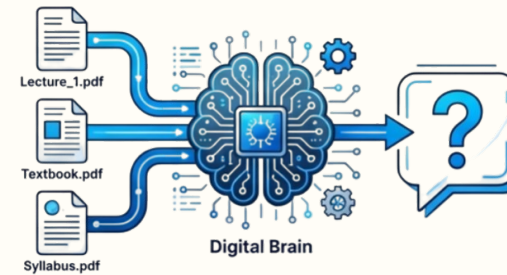
The old way: Context-unaware direct answers



Generic LLMs like ChatGPT, gemini, etc lack course specific context and provide direct answers, bypassing the learning process.

Direct answers, minimal learning

Our solution: Context-bound Socratic tutoring



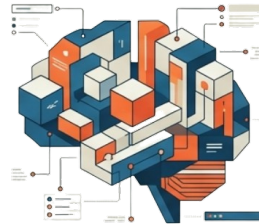
Context-Bound RAG Engine. The system doesn't provide answers directly. Instead, it guides the students to the answer via follow-up questions and tutoring.

Passive learning, beyond classroom!



Context-Bound

Restricted strictly to uploaded Data. Refuses out-of-scope queries.



Socratic Engine

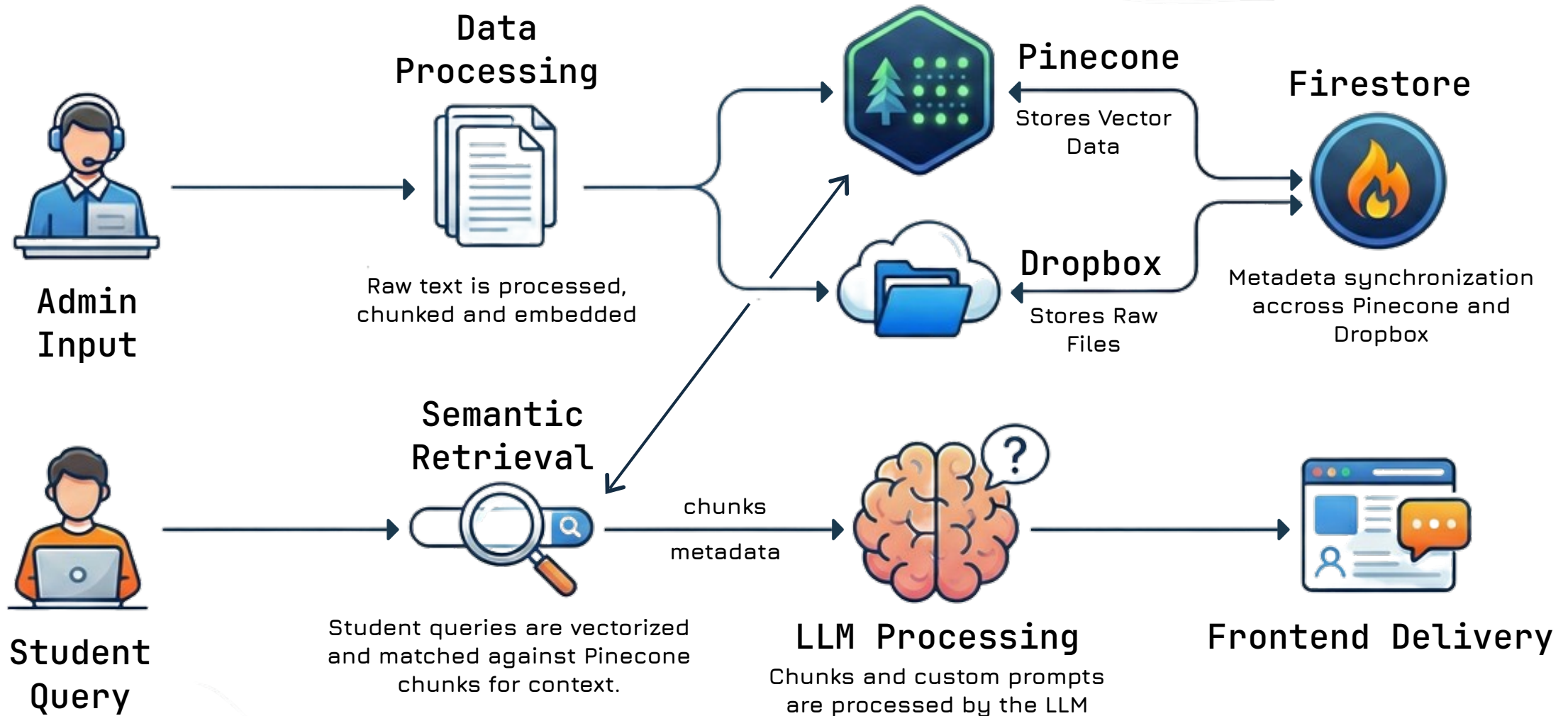
AI guides students via follow-up questions, not direct answers.



Faculty Insights

Admin Dashboard with "Confusion Heatmaps" to identify struggle areas.

ARCHITECTURE DIAGRAM



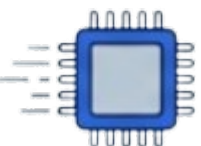
FEASIBILITY & VIABILITY

Implementation Feasibility

- ◆ **Serverless Strategy:** Leveraging vercel Deployment and Pinecone eliminates infrastructure management.
- ◆ **Pre-trained Models:** Employing Groq/Gemini API bypasses model training times and overheads.
- ◆ **Modular Development:** Decoupled Backend (Python) and Frontend (React) allows parallel team workflows.



Challenges vs Solutions

	Insufficient or missing contextual data can cause the model to generate fabricated responses, breaking the course-aware behavior of the system.		Differences in how students phrase questions can lead to inconsistent or weak document retrieval, reducing answer relevance.		Lack of reliable linkage between responses and their originating document sections reduces clarity and ability to access relevant sources.
	Retrieval-gated prompting restricts generation strictly to available course content and enforces refusal when context is missing.		Optimized chunking with high-quality embeddings and metadata filtering stabilizes semantic retrieval across query variations.		Page-level metadata and citation-aware generation ensure answers remain grounded with sources.

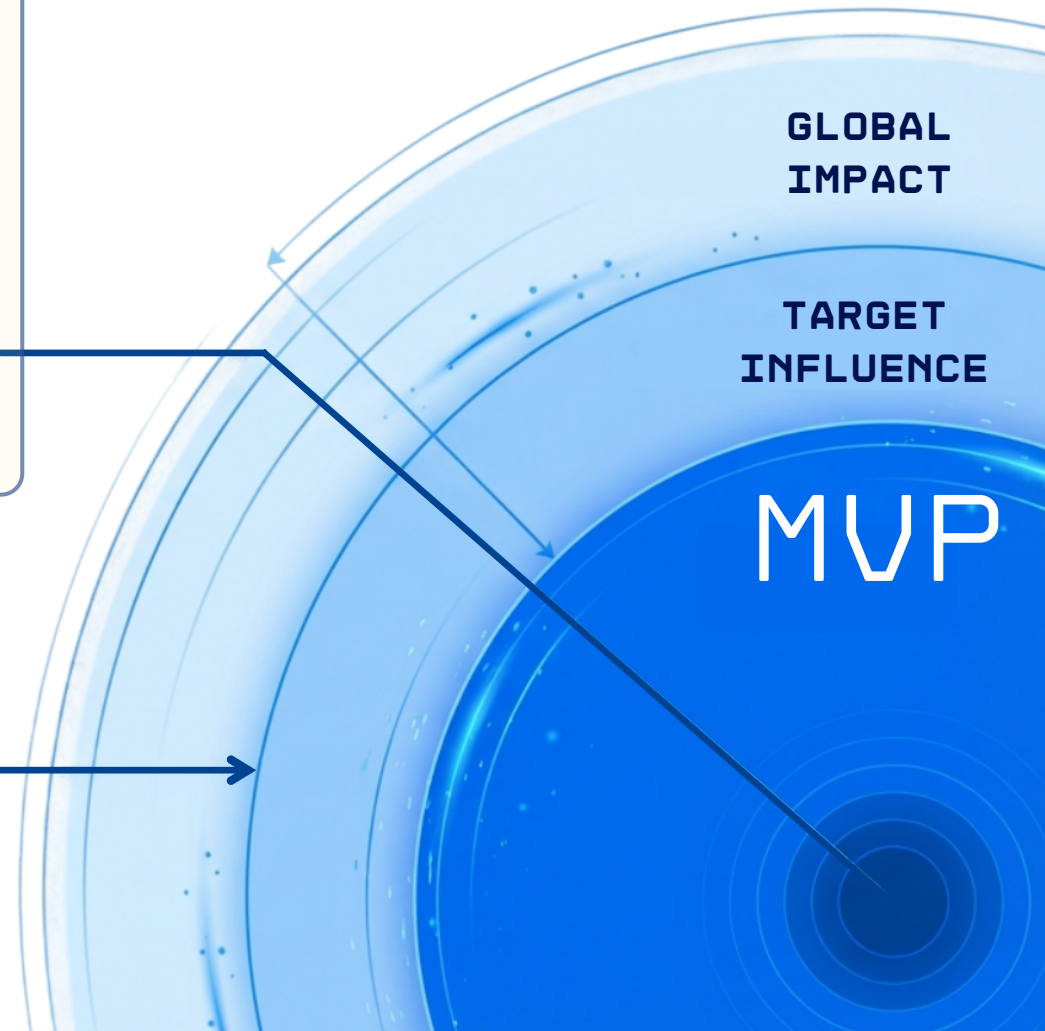
IMPACT AND BENEFITS

Immediate Utility (MVP)

- ✓ Real-time, course-specific knowledge base built directly from uploaded lecture notes and textbooks.
- ✓ Faculty-controlled document management with live add, delete, and archive capabilities.
- ✓ Interactive, low-latency tutoring grounded strictly in uploaded material.
- ✓ Admin analytics dashboard highlighting confusing topics and student-wise engagement using heatmap-based insights

Global Impact and Scale

- ✓ Multi-tenant architecture enabling secure deployment across universities worldwide
- ✓ High-quality tutoring access comparable to top-tier institutions through locally deployed, course-trained models
- ✓ Continuous curriculum refinement driven by aggregated learning analytics and student performance trends



RESEARCH AND REFERENCES

Our solution is built upon established research in [Retrieval-Augmented Generation \(RAG\)](#), [intelligent tutoring systems](#), and [learning analytics](#).

KEY RESEARCH AREAS

Retrieval-Augmented Generation (RAG)

Combines the strengths of information retrieval and generative models to produce accurate, context-grounded responses.

LEWIS ET AL., 2020

Socratic Teaching Methods

Engaging learners through guided questioning improves critical thinking and deeper conceptual understanding.

HMELO-SILVER, 2004

Contextual & Secure Information Retrieval

Contextual filtering and access control ensure responses are grounded in trusted, institutional content.

GUU ET AL., 2020

Learning Analytics & Insight Generation

Analytics dashboards and heatmaps help educators identify learning gaps and optimize interventions.

SIEMENS & LONG, 2011

REFERENCES



Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis et al., 2020

NeurIPS 2020 — Introduces RAG for factual consistency by grounding generation in retrieved documents.



Problem-Based Learning: What and How Do Students Learn?

Catherine Hmelo-Silver, 2004

Educational Psychology Review — Guided inquiry and questioning enhance deep learning.



GPT-4 Technical Report

OpenAI, 2023

Details the architecture and capabilities of GPT-4, the LLM powering our tutoring engine.



Pinecone Documentation

Pinecone, 2024

Official documentation for scalable vector databases used for semantic search and retrieval.



Firebase Documentation

Google, 2024

Reference for Firestore usage in metadata storage, authentication, and real-time sync.